Mammalia Raccoon Proximity Network Analysis

https://github.com/aiyshwary/Raccoon-Network-Analysis

Python NetworkX Jupyter Graph Analysis Epidemiology

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

TECHNICAL WALKTHROUGH

Mammalia Raccoon Proximity Network Analysis – Project

Explanation

Project Overview
This project analyzes social interaction patterns among urban raccoons using network analysis to understand how rabies and other pathogens can spread within raccoon populations.
Since raccoons are a major vector of rabies, studying how often and how closely they interact helps estimate disease transmission risk and supports better public health and wildlife management strategies.

Objective

Model raccoon interactions as a graph/network
Identify high-risk raccoons that can spread pathogens quickly Use centrality measures to understand social structure and disease
flow

Dataset & Network Construction
24 raccoons → represented as nodes
~2000 interactions → represented as weighted edges

Data collected using proximity-detecting collars
Edge weight = time spent in close proximity Monthly interaction data used to build the network

If two raccoons are connected, it means they had close physical
interaction, which is critical for rabies transmission.
Why Network Analysis?
Disease spread is not random.

It depends on

1. Who interacts with whom

2. How frequently

3. Who acts as a bridge between groups
4. Network analysis helps identify super-spreaders and critical connectors.
5. Centrality Measures Used & Interpretation
6. Degree Centrality
7. Measures number of direct connections
8. High degree = more interactions

Inference

1. Raccoon #4 has the highest degree
2. Acts as a high-contact individual
3. High risk of initiating widespread infection
4. Closeness Centrality
5. Measures how close a node is to all others
6. Indicates how fast something can spread

Inference

1. Raccoon #2 has highest closeness centrality
2. Can spread pathogens rapidly across the entire network
3. Betweenness Centrality
4. Measures how often a node lies on shortest paths
5. Identifies bridge nodes

Inference

1. Raccoon #16 connects different subgroups
2. Acts as a key transmission bridge
3. Removing or vaccinating this raccoon can significantly slow disease
4. spread
5. Clustering Coefficient
6. Measures how tightly a raccoon is embedded in a local group
7. Indicates family or group behavior

Inference

1. Raccoon #2 has highest clustering
2. Infection here may cause localized outbreaks within families
3. Eigenvector Centrality
4. Measures influence based on connections to influential nodes

Inference

1. Raccoon #4 again ranks highest
2. Connected to other highly interactive raccoons
3. Considered a super-spreader candidate
4. PageRank Algorithm
5. Evaluates overall importance considering direction and weight of
6. interactions

7. Helps rank raccoons by global influence
8. Key Findings
9. Raccoon populations are highly interconnected
10. Disease can spread much faster than expected
11. A few raccoons play disproportionately large roles in pathogen
12. transmission
13. Targeted intervention (vaccination/tracking of key raccoons) is more
14. effective than random control
15. Technologies & Concepts Used
16. Graph Theory
17. NetworkX (Python)
18. Weighted graphs
19. Centrality algorithms
20. Epidemiological modeling concepts

KEY DETAILS

1. Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.
2. Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.
3. Identified superspreader nodes whose removal would most disrupt disease propagation chains.

4. Visualized network topology and centrality distributions to communicate findings clearly.

5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.